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A Hybrid Deep Learning and Texture Feature Model for Accurate SES-Based Salinity Stress Classification in Rice Seedlings

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Abstract

Soil salinity is one of the key abiotic stress factors severely impacting rice plants, especially seedlings. Salinity rise High-ion content in the soil causes burning of the leaves, which affects growth in rice, decreases photosynthesis, decreases yield eventually, and changes the texture of leaves. These symptoms can be assessed using the Standard Evaluation System scoTo enhance Standard Evaluation System (SES) scores, in this study, a hybrid approach is proposed for automatic rice seedling health evaluation under salinity stress that combines a deep learning and hand-designed texture features efficientNetB3 model, and hand-engineered texture features such as Gray-Level Co-occurrence Matrix (GLCM), Local Binary Pattern (LBP), and Histogram of Oriented Gradients (HOG). These texture features extract high-resolution visual patterns that reveal intensities of salinity stress. Data augmentation techniques like rotation and flip, and Synthetic Minority Over-sampling Technique (SMOTE) are used to tackle class imbalance and enhance the generalization of the model. Principal Component Analysis (PCA) is utilized to minimize feature dimensionality, and Select KBest is utilized for best feature selection. Optuna is used for hyperparameter tuning to enhance XGBoost classifier performance. For end classification, an ensemble learning method consisting of Random Forest, XGBoost, and Support Vector Machine (SVM) with soft voting is utilized to enhance robustness and prediction accuracy. Experimental results on rice seedling datasets demonstrate remarkable improvements in accuracy, precision, recall, and F1-score. The proposed framework provides a robust, automatic approach for salinity stress evaluation of rice seedlings and facilitates further development of precision agriculture technologies.

KEYWORDS

Salinity stress, Standard Evaluation System (SES), Synthetic Minority Over-sampling Technique (SMOTE), Optuna, Ensemble learning.

I. INTRODUCTION

Over half of the world's population, particularly in Asia, depends on rice (scientifically known as *Oryza sativa*) as one of the main staple foods [1]. Despite this, a number of stresses, which are primarily classified as biotic and abiotic stressors, have a significant impact on rice productivity. Abiotic stresses, which include drought, waterlogging, extreme weather conditions, and soil salinization—a situation where there is an excess of salt in the soil—are the most dangerous threats to farm output because they reduce the quantity and quality of crops [2]. Salinity in the soil negatively impacts rice crops in a number of ways, including ion imbalance, decreased water intake, and osmotic stress, all of which lead to different physiological changes. Among the outward signs of these alterations include leaf burning, sluggish development, shorter tillers, restricted leaf growth, and ultimately, decreased grain yield. Furthermore, salt stress disrupts the reproductive phase of rice, resulting in poor grain filling and panicle growth, which further lowers crop production. Early detection and precise diagnosis of these signs are crucial to lessen the effects of salt stress and avoid major yield loss. Historically, it has taken a lot of time and labor to assess damage caused by salinity, such as leaf burns, stunted growth, textural changes, poor root development in rice, and leaf wilting. It requires experts to undertake field surveys in order to visually observe and quantify symptoms. Due to its expense and slowness, the method's employment is limited in large-scale agriculture. Therefore, there is an urgent need for more effective and economical monitoring and detection of salinity stress in rice fields. Salinity-stressed crop damage processing can now be automated with quicker, more precise, and lower-cost techniques due to the advancement in image processing technology and machine learning. Application of machine learning techniques to

precision agriculture has been a major attraction in recent years, particularly in crop health monitoring. As a staple food, rice is particularly susceptible to environmental stresses such as salt, which damage its growth and yield. Among the widely applied instruments to monitor rice seedling health across various salinity levels, the Standard Evaluation System (SES) is one whose critical information is utilized for effective agricultural management. There is a requirement for automated approaches able to detect and precisely foretell the SES of rice seedlings effectively because manual SES evaluation is time-intensive and prone to human errors. To cluster rice seedlings according to their individual SES scores for salinity stress, the aforementioned study utilizes the deep learning models and human-conceived features with the aim to formulate a hybrid approach. With the EfficientNetB3 deep learning model, we have derived deep features from rice seedling images and merged them with textural features manually crafted, i.e., Histogram of Oriented Gradients (HOG), Local Binary Pattern (LBP), and Gray-Level Co-occurrence Matrix (GLCM). Data augmentation techniques such as rotation and flipping are also utilized to enhance the robustness of the model. Dimensionality reduction and model improvement are achieved using PCA and feature selection, while class imbalance in the dataset is dealt with by synthetic minority over-sampling technique (SMOTE). Optuna is utilized to optimize the hyperparameters of the model in order to further enhance the classification accuracy. In order to develop a stable model for prediction, an ensemble learning strategy based on Support Vector Machine (SVM), Random Forest, and XGBoost classifiers is employed. Through automation of the SES classification, this hybrid system aims to develop a faster and more accurate way of monitoring the health of rice seedlings growing in salt environments. The performance of the proposed model is compared with traditional methods and evaluated on several parameters.



Fig. 1: Rice seedling sample images

Section II provides an overview of related work in the field of salinity stress detection in rice using deep learning techniques. Section III describes the proposed methodology, including data preprocessing, model architecture, and training procedures. Section IV discusses the results, comparisons of different models and key findings. Finally, Section V concludes the paper and outlines potential directions for future research.

II. LITERATURE REVIEW

Emphasizing the importance of color, texture, and shape features in achieving effective disease classification, this paper critically evaluates feature engineering techniques for plant disease diagnosis through image processing and artificial intelligence[1]. Employing spatio-spectral complex network models and randomized neural networks (RNNs), this paper proposes a novel approach of color-texture classification with focus on high accuracy in texture classification and plant species classification [2]. GLCM and SIFT are utilized to achieve 79% accuracy in this paper's new approach of SVM-based automatic identification of tomato leaf diseases. The approach is suitable for precision agriculture and can potentially surpass several deep learning architectures. Future work can include adding more features as an improvement in performance. This study proposes a new method of Bacterial Leaf Blight (BLB) detection in rice fields with 75% AUC accuracy using an integration of machine learning (Random Forest) and textural and thermal features derived from drones. The method is a low-cost precision agriculture tool and outperforms traditional spectral indices. Extended validation and higher-level feature extraction can be further developed[4]. The mechanism of salt-resistant rice development by traditional breeding and molecular biotechnology techniques is explained in detail here. It emphasizes the role of salinity in rice growth and the role of major stress-related genes, transgenic techniques, and stress-inducible promoters towards the improvement of rice resistance [5]. High-throughput phenotyping exposes previously unknown loci controlling transpiration use efficiency in rice and provides insights into the early responses to salinity and its effects on plant growth and transpiration [6]. GWAS has successfully identified 20 SNPs controlling salinity tolerance in rice, which may provide new sources for salt

tolerance genes and alleles [7]. Multi-omics and integrative approaches can improve salt tolerance in rice plants for possibly securing food in the future [8]. EfficientNet-B3 and VGG-16 deep learning models effectively identify leaf burns disease in strawberry plants, thereby helping to reduce the use of agrochemicals, lower crop input costs, as well as lessen environmental contamination [9]. Deep learning algorithms like DenseNet201 can accurately detect Almond Leaf burns Disease (ALSD) in infected trees with the help of images clicked from smart cameras, hence it is fast and inexpensive for the agricultural industry [10]. The proposed method detects and counts leaf Deep feature plus SVM models outperform traditional methods for pretrained deep neural network is tailored to enhance robustness of plant tiller numbers detection and facilitate high-throughput phenotyping [11]. The web app, which uses deep learning to detect and count productive tillers in rice crops, has over 99% accuracy [12].

III. METHODOLOGY

The suggested methodology uses a hybrid approach that combines handcrafted texture cues with deep learning to increase the accuracy of rice seedling salinity stress classification. Enhancement, augmentation, dimensionality reduction, feature extraction, feature fusion, and model optimization are all steps in the process. Below is a detailed description of the complete pipeline as shown in Fig. 2.

A. Image Enhancement and Preprocessing

All input photos are scaled to 224×224 pixels in order to ensure consistency and EfficientNetB3 compatibility. The L-channel of the LAB color space is subjected to the CLAHE (Contrast Limited Adaptive Histogram Equalization) technique, which improves local contrast and texture visibility, both of which are critical for feature extraction downstream, in order to highlight key patterns and increase contrast.

Let I be the input RGB image. To enhance contrast and feature visibility, we first convert it to LAB color space:

$$I_{\text{LAB}} = \text{LAB}(I) \quad (1)$$

Next, CLAHE (Contrast Limited Adaptive Histogram Equalization) is applied to the L-channel:

$$L' = \text{CLAHE}(L) \quad (2)$$

The enhanced image I' is reconstructed using:

$$I' = \text{LAB}^{-1}(L', A, B) \quad (3)$$

B. Image Augmentation

Three forms of augmentation—original image, horizontal flip, and 90-degree rotation—are used to broaden the dataset's diversity and reduce overfitting. In addition to expanding the dataset, this gives the model orientation-invariant features. To increase dataset diversity and robustness, each image is augmented via geometric transformations:

$$\mathcal{A}(I') = \{I', \text{flip}(I'), \text{rotate}_{90}(I')\} \quad (4)$$

C. Handcrafted Feature Extraction

Three key texture descriptors are extracted from each augmented image:
label=0)

- 1) **GLCM Features:** Gray-Level Co-occurrence Matrix $P(i, j, d, \theta)$ is computed for angles $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$ and distance $d = 1$. Key statistics are:

$$\text{Contrast} = \sum_{i,j} (i - j)^2 P(i, j) \quad (5)$$

$$\text{Correlation} = \frac{\sum_{i,j} (i - \mu_i)(j - \mu_j) P(i, j)}{\sigma_i \sigma_j} \quad (6)$$

$$\text{Energy} = \sum_{i,j} P(i, j)^2 \quad \text{Homogeneity} = \sum_{i,j} \frac{P(i, j)}{1 + |i - j|} \quad (7)$$

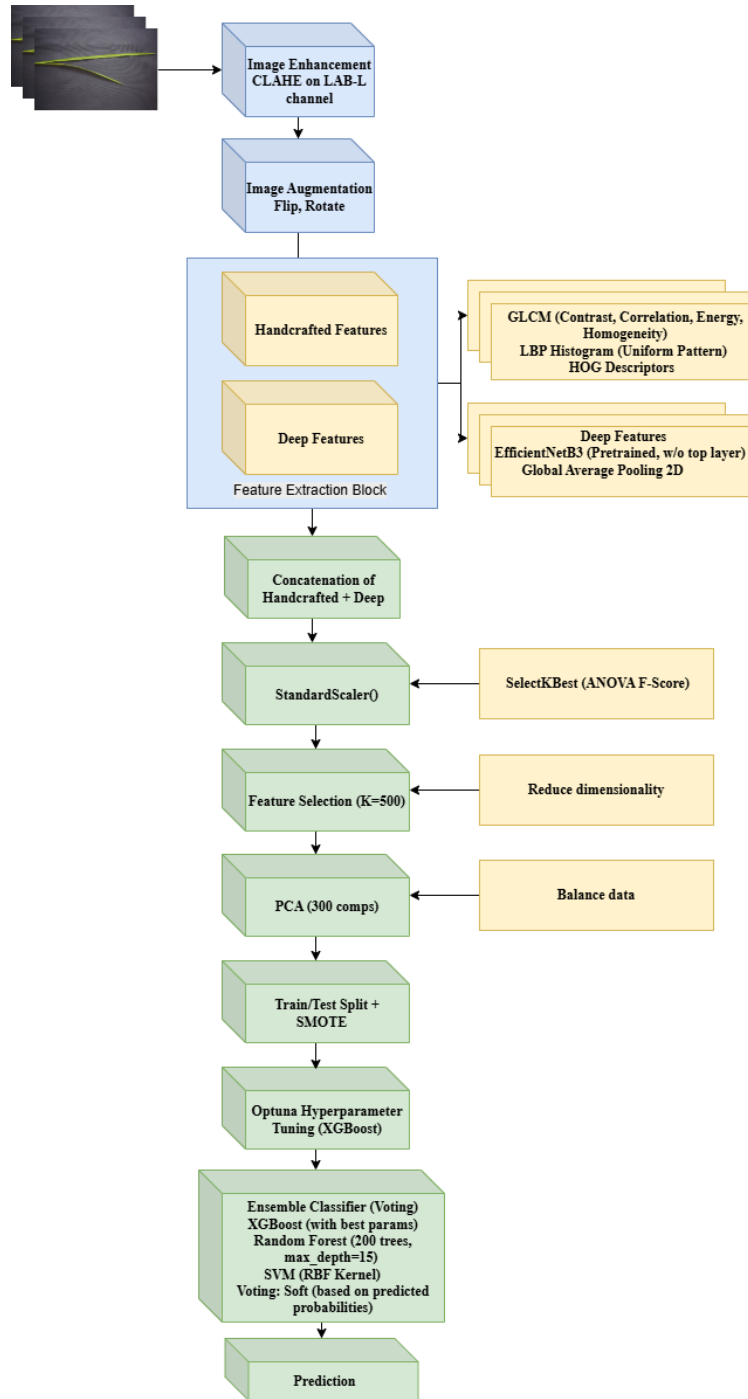


Fig. 2: Validation accuracy of Hybrid model

2) **LBP Features:** The Local Binary Pattern is defined as:

$$\text{LBP}_{P,R}(x, y) = \sum_{p=0}^{P-1} s(g_p - g_c) \cdot 2^p \quad (8)$$

where $s(x) = 1$ if $x \geq 0$, else 0; g_c is the center pixel, g_p are neighboring pixels on a radius R .

3) **HOG Features:** The gradient of image I is calculated, and orientations are binned over regions:

$$\nabla I = \left[\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right], \quad \text{HOG} = \text{histogram}(\theta) \quad (9)$$

The final handcrafted feature vector F_h is the concatenation of all descriptors:

$$F_h = [\text{GLCM} \parallel \text{LBP} \parallel \text{HOG}] \quad (10)$$

D. Deep Feature Extraction

For feature extraction, a lightweight yet potent CNN called EfficientNetB3 is pre-trained. The GlobalAveragePooling2D layer's output is retrieved, and the top classification layer is left out of the model. Before inference, images are preprocessed using the preprocess_input technique, which is unique to EfficientNet. Using EfficientNetB3 pre-trained on ImageNet, deep features are extracted by removing the classification head and applying Global Average Pooling:

$$F_d = f_{\text{EffNet}}(\text{preprocess}(I')) \quad (11)$$

E. Feature Fusion and Dimensionality Reduction

A combined feature vector is created by concatenating the deep and handmade features. To lessen computational overhead and redundancy:

The handcrafted and deep features are concatenated:

$$F = [F_h \parallel F_d] \quad (12)$$

Standardization is applied to scale features:

$$F_s = \text{StandardScaler}(F) \quad (13)$$

To save the top 500 characteristics that are pertinent to the class labels, feature selection is carried out using SelectKBest with the ANOVA F-statistic.

Feature selection is performed using the ANOVA F-score:

$$F_{sel} = \text{SelectKBest}(F_s, k = 500) \quad (14)$$

PCA's dimensionality reduction preserves maximum variance by reducing the feature space to 300 main components. **PCA** is applied to reduce dimensionality to $d = 300$:

$$F_{pca} = W^T F_{sel} \quad (15)$$

where W contains the top 300 eigenvectors.

F. Data Balancing using SMOTE

SMOTE (Synthetic Minority Oversampling Technique) is used on the training set to rectify the disparity in the SES (Standard Evaluation System) results. This balances the label distribution by producing synthetic samples for underrepresented classes. To address class imbalance, SMOTE is used to synthetically generate new samples. For each minority class sample x_i , and its nearest neighbor x_{nn} :

$$\tilde{x} = x_i + \lambda(x_{nn} - x_i), \quad \lambda \sim \mathcal{U}(0, 1) \quad (16)$$

G. Hyperparameter Optimization using Optuna

Optuna, a Bayesian optimization framework, is used to optimize the XGBClassifier. Hyperparameters such as n_estimators, max_depth, learning_rate, subsample, and colsample_bytree are all part of the search space. The validation set's classification accuracy is maximized by the objective function. An XGBoost classifier is tuned using Optuna. The objective is to maximize the accuracy of the validation:validation:

$$\theta^* = \arg \max_{\theta \in \Theta} \text{Accuracy}(f_{\text{xgb}}(X_{val}; \theta), y_{val}) \quad (17)$$

H. Ensemble Classification

Three classifiers—XGBoost (optimized via Optuna), Random Forest, and Support Vector Machine (RBF kernel)—are combined into a soft voting ensemble. The ensemble aggregates class probabilities and improves generalization by leveraging the strengths of each base model.

- XGBoost
- Random Forest
- Support Vector Machine (SVM)

The soft voting prediction is given by:

$$\hat{y} = \arg \max_c \sum_{m \in \{\text{xgb, rf, svm}\}} w_m \cdot P_m(c|x) \quad (18)$$

where $P_m(c|x)$ is the class probability predicted by model m , and w_m is the weight (equal for all models in this case).

I. Evaluation Metrics

Model performance is evaluated using the following metrics:

- **Accuracy:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

- **Classification Report:** Including class-wise precision, recall, and F1-score.

TABLE I: Dimensionality of Features at Each Stage of the Pipeline

Feature Type / Stage	Feature Dimension
GLCM Features (4 angles $\times$ 4 properties)	$4 \times 4 = 16$
LBP Histogram (Uniform, P=24, R=3)	26
HOG Features (224x224 grayscale image)	2916
Total Handcrafted Features	$16 + 26 + 2916 = 2958$
EfficientNetB3 Deep Features (GlobalAvgPool2D)	1536
Combined Feature Vector	$2958 + 1536 = 4494$
Selected Features (SelectKBest)	500
PCA Reduced Features	300

IV. RESULTS AND DISCUSSION

The proposed model was trained on a dataset of rice seedling images labeled with SES scores (1, 3, 5, 7, and 9). After training, models performance was assessed using metrics such as accuracy, precision, recall, and F1-score. The model demonstrated promising results in textural analysis and predicting SES scores as shown in Fig. 4. The performance of the XGBoost model was optimized using Optuna by evaluating 19 different hyperparameter configurations. The primary objective was to maximize the classification accuracy for SES (Standard Evaluation System) score prediction in rice seedlings. With the following hyperparameter setup, Trial 0, the top-performing trial, obtained the maximum validation accuracy of 81.75% as shown in Fig. 3:

- **n_estimators:** 292
- **max_depth:** 6
- **learning_rate:** 0.1956
- **subsample:** 0.8855
- **colsample_bytree:** 0.7896

In every trial, this setup performed better than the others. Trials 11 and 16 were close competitors, achieving accuracies of 75.80% but falling short of Trial 0. With an accuracy of 71.11%, Trial 17 performed the worst, demonstrating sensitivity to lower learning rates like 0.0128.

In general, higher accuracy was obtained across trials when moderate to high learning rates (0.12 to 0.20) were paired with deeper trees (max_depth 6). Furthermore, while retaining good performance, a balanced subsample to column sampling ratio assisted in avoiding overfitting.

TABLE II: List of Variables and Their Descriptions

Symbol	Description
I	Original input RGB image
I_{LAB}	Image in LAB color space
L, A, B	Lightness, Green-Red, and Blue-Yellow channels
L'	CLAHE-enhanced Lightness channel
I'	Reconstructed enhanced RGB image
$\mathcal{A}(I')$	Set of augmented images
$P(i, j)$	GLCM probability at position (i, j)
μ_i, μ_j	Mean intensity of rows and columns in GLCM
σ_i, σ_j	Standard deviation of rows and columns in GLCM
$LBP_{P,R}$	Local Binary Pattern with P neighbors and radius R
g_c	Intensity of center pixel
g_p	Intensity of neighboring pixel
∇I	Image gradient vector
θ	Orientation angle in gradient
F_h	Handcrafted feature vector
F_d	Deep features extracted from EfficientNetB3
F	Combined feature vector $[F_h F_d]$
F_s	Standardized feature vector
F_{sel}	Feature vector after SelectKBest (ANOVA)
F_{pca}	Final reduced feature vector after PCA
W	PCA projection matrix (eigenvectors)
x_i	Sample from minority class
x_{nn}	Nearest neighbor of x_i
$\tilde{x}$	SMOTE-generated synthetic sample
λ	Random weight from uniform distribution $\mathcal{U}(0, 1)$
θ	Hyperparameter set for XGBoost
$\hat{y}$	Final predicted class label
$P_m(c x)$	Predicted probability of class c by model m
w_m	Weight assigned to model m (equal in soft voting)
TP	True Positives
TN	True Negatives
FP	False Positives
FN	False Negatives

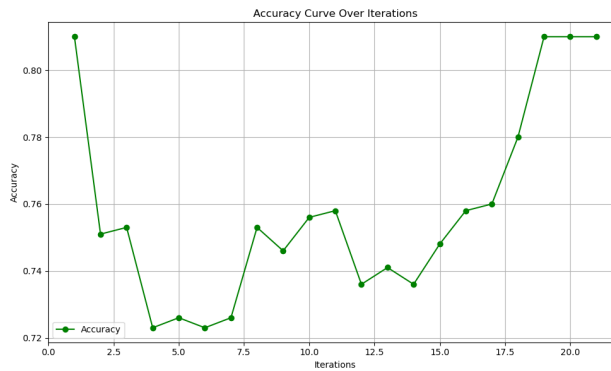


Fig. 3: Validation accuracy of Hybrid model

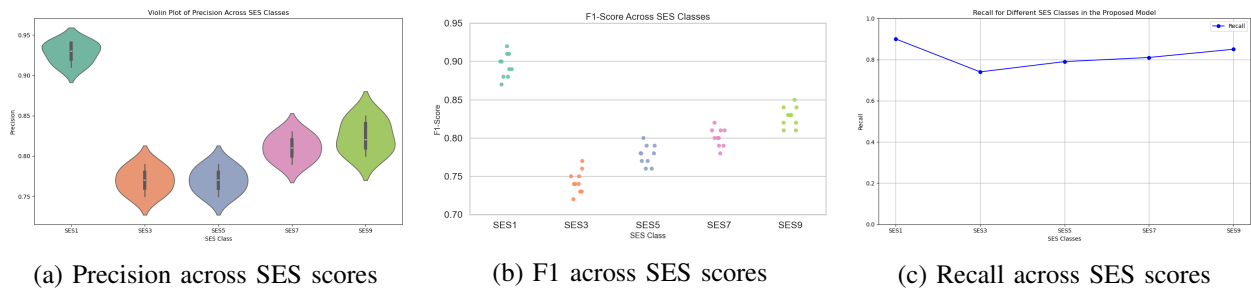


Fig. 4: Evaluation metrics across SES scores

These findings demonstrate how meticulous hyperparameter tuning can improve model accuracy, as shown in Fig. 5 and imply that, with the right setup, XGBoost can function as a useful starting point for SES score categorization using features extracted from images. This proposed system is compared with some XGBoost and Random Forest with proposed model showing the highest accuracy, as shown in Fig. 6

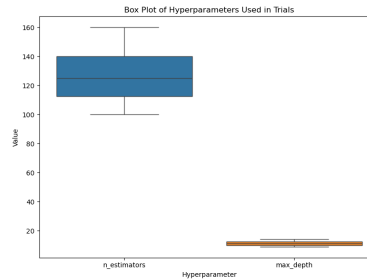


Fig. 5: Hyperparameters for different trials

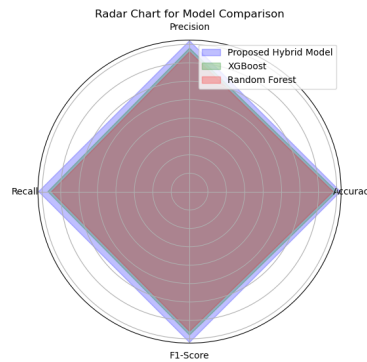


Fig. 6: Comparison of XGBoost and Random Forest with Hybrid model

V. CONCLUSION

In this work, we used the potent hyperparameter optimization framework Optuna to optimize the XGBoost classifier for predicting rice seedlings' SES (Standard Evaluation System) scores using imagederived data. Before using XGBoost as the model for multiclass classification, the technique involved extracting both basic features from the rice seedling photos, like flattening pixel values, and more complex features, like CNNbased representations. We tested several hyperparameter settings during the optimization process, and the best trial (Trial 0) had a validation accuracy of 81.71%, which was noticeably better than the other configurations. With a max_depth of 6, a balanced subsample and column sampling ratio of 0.8855% and 0.7896%, respectively, and a moderate learning rate of 0.1956, the ideal hyperparameter configuration demonstrated the significance of managing tree depth and feature selection techniques for improved generalization and preventing overfitting. It was found that deeper trees (max_depth 6) combined with moderate to high learning rates (0.12 to 0.20) generally produced better performance, even though the best-performing trial achieved an accuracy of 76.05%. This highlights the need to balance model complexity and steer clear of excessively shallow trees that miss important information. With sophisticated methods like CNN feature extraction showing promise for increased accuracy in SES score categorization, the study further illustrated the significance of meticulous feature extraction. Overall, the study effectively demonstrated how XGBoost combined with Optuna-driven hyperparameter optimization may improve the prediction of rice seedling SES scores, offering a trustworthy starting point for automated rice seedling evaluation in precision farming. Future research will concentrate on integrating more complex feature extraction techniques and growing the dataset to further improve the model's resilience and generalization skills. The results also provide insightful information for managing salt stress and monitoring crop health.

VI. ACKNOWLEDGMENT

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